# Lección 2

En Debian,

→ ssh -p 22022 ignaciordt@192.168.56.105

→ cat /proc/mdstat → veo que

RAID1 [2/2]: están activados y

[UU] ninguno ha fallado.

→ su -

→ nano mon_raid.py y copiamos y pegamos de los ficheros descargados de prado.

└ Comprueba si algún disco pasa a [U_] o [_U].

```
ignaciordt@debian-ise:~$ cat /proc/mdstat
Personalities : [raid1] [raid0] [raid6] [raid5] [raid4] [raid10]
md0 : active raid1 sdb1[2] sda1[0]
      486400 blocks super 1.2 [2/2] [UU]

md1 : active raid1 sdb2[2] sda2[0]
      9987072 blocks super 1.2 [2/2] [UU]

unused devices: <none>
ignaciordt@debian-ise:~$ nano ~/mon_raid.py
ignaciordt@debian-ise:~$ python3 ~/mon_raid.py
--All RAIDs OK--
--OK Script--
ignaciordt@debian-ise:~$
```

→ cd /etc/systemd/system Ejecuta mon_raid.py Cada minuto se ejecuta. Activa Mon.service cada minuto.

→ nano mon_raid.service y nano mon_raid.timer

y copiamos y pegamos de los ficheros descargados de prado.

→ systemctl daemon-reload recargo systemd

→ systemctl start mon_raid arranco

→ systemctl status mon_raid

```
root@debian-ise:~# systemctl start mon_raid
systemctl status mon_raid
● mon_raid.service - Monitor RAID status
     Loaded: loaded (/etc/systemd/system/mon_raid.service; static)
     Active: active (running) since Sun 2026-11-08 21:12:40 CET; 17ms ago
 Invocation: 9a6eb3eaa05344f5b62aa1e3e50ac1c3
TriggeredBy: ○ mon_raid.timer
   Main PID: 6622 ((python3))
      Tasks: 1 (limit: 2297)
     Memory: 1.5M (peak: 1.7M)
        CPU: 3ms
     CGroup: /system.slice/mon_raid.service
             └─6622 "(python3)"

nov 08 21:12:40 debian-ise systemd[1]: Started mon_raid.service - Monitor RAID
lines 1-13/13 (END)...skipping...
● mon_raid.service - Monitor RAID status
```

¡Captura Github!

→ systemctl daemon-reload recargo systemd

→ systemctl start mon_raid.timer arranco

→ systemctl enable mon_raid.timer

└ Hago enable solo en timer porque queremos que el servicio se ejecute automáticamente cada cierto tiempo, no que se lanza por sí solo al arrancar Debian.

```
root@debian-ise:~# systemctl daemon-reload
root@debian-ise:~# systemctl start mon_raid.timer
systemctl enable mon_raid.timer
Created symlink '/etc/systemd/system/timers.target.wants/mon_raid.timer' → '/etc
/systemd/system/mon_raid.timer'.
root@debian-ise:~# systemctl status mon_raid.timer
● mon_raid.timer - Temporizador para monitorizar el RAID
     Loaded: loaded (/etc/systemd/system/mon_raid.timer; enabled; preset: enabl
     Active: active (waiting) since Sun 2026-11-08 20:39:17 CET; 5s ago
 Invocation: e267bad21086412a8362474295fcf764
    Trigger: Sun 2026-11-08 20:40:17 CET; 54s left
   Triggers: ● mon_raid.service
```

¡Captura Github!

tiempo, no que se lanza por sí solo al arrancar Debian.

¡Captura Github!

→ systemctl status mon_raid.timer

→ journalctl -u mon_raid.service

Consulta los registros del sistema — Muestra los mensajes de nuestro servicio

```
nov 08 20:39:17 debian-ise python3[5804]: --All RAIDs OK--
nov 08 20:39:17 debian-ise python3[5804]: --OK Script--
nov 08 20:39:17 debian-ise systemd[1]: mon_raid.service: Deactivated successfully.
nov 08 20:39:17 debian-ise systemd[1]: Finished mon_raid.service - Monitor RAID
nov 08 20:40:27 debian-ise systemd[1]: Starting mon_raid.service - Monitor RAID
nov 08 20:40:27 debian-ise python3[5846]: --All RAIDs OK--
nov 08 20:40:27 debian-ise python3[5846]: --OK Script--
nov 08 20:40:27 debian-ise systemd[1]: mon_raid.service: Deactivated successfully.
nov 08 20:40:27 debian-ise systemd[1]: Finished mon_raid.service - Monitor RAID
nov 08 20:41:37 debian-ise systemd[1]: Starting mon_raid.service - Monitor RAID
nov 08 20:41:37 debian-ise python3[5857]: --All RAIDs OK--
nov 08 20:41:37 debian-ise python3[5857]: --OK Script--
lines 1-12
```

→ journalctl -u ssh --since="yesterday"

└ Muestra las conexiones ssh registradas desde ayer.

```
root@debian-ise:~# journalctl -u ssh --since="yesterday"
nov 08 21:09:32 debian-ise sshd-session[6481]: Accepted password for ignaciordt
 from 192.168.56.1 port 58240 ssh2
nov 08 21:09:32 debian-ise sshd-session[6481]: pam_unix(sshd:session): session
opened for user ignaciordt(uid=1000) by ignaciordt(uid=0)
root@debian-ise:~#
```

# Ejercicio 2

→ apt update

→ apt install ansible → ansible es una herramienta que permite administrar y ejecutar comandos en varios servidores desde 1 sola máquina

→ exit salimos de root

→ nano hosts

Si no pongo el puerto 22022 no me deja conectarme con ansible

```
GNU nano 8.4                    hosts
192.168.56.105 ansible_port=22022  Debian
192.168.56.102 ansible_port=22022  Alma
```

Aparece este error →



→ ansible -i hosts all -m ping -u ignaciordt

ansible : ejecuta ansible para administrar servidores.

-i hosts : utiliza el archivo host como inventario de máquinas.

all : ejecuta la operación en todas las máquinas del inventario.

-m ping: para comprobar la conexión.

-u ignaciordt: se conecta al usuario ignaciordt.

```
ignaciordt@debian-ise:~$ ansible -i hosts all -m ping -u ignaciordt
[ERROR]: Task failed: Failed to connect to the host via ssh: Host key verification failed.
Origin: <adhoc 'ping' task>

{'action': 'ping', 'args': {}, 'timeout': 0, 'async_val': 0, 'poll': 15}

192.168.56.105 | UNREACHABLE! => {
    "changed": false,
    "msg": "Task failed: Failed to connect to the host via ssh: Host key verification failed",
    "unreachable": true
}
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/pyth
preter_discovery.html for more information.
192.168.56.102 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "changed": false,
    "ping": "pong"
}
ignaciordt@debian-ise:~$
```

Por eso cuando desde Debian me intento conectar a Alma

NO me pide contraseña

```
[ignaciordt@debian-ise:~$ ssh -p 22022 ignaciordt@192.168.56.102
Last login: Mon Nov  2 15:34:35 2020 from 192.168.56.105
[ignaciordt@bbox ~]$
```

Pero de Alma a Debian si me la pide.

```
[ignaciordt@bbox ~]$ ssh -p 22022 ignaciordt@192.168.56.105
The authenticity of host '[192.168.56.105]:22022 ([192.168.56.105]:22022)' can'
t be established.
ED25519 key fingerprint is SHA256:LbSzeeINvjCcGPYtlvfbZBsTT3D4iCSciUiVQnKzeHsU2A.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[192.168.56.105]:22022' (ED25519) to the list of kn
own hosts.
ignaciordt@192.168.56.105's password:
```

↳ A Alma se puede conectar y a Debian no, esto es porque en Alma tiene nuestra llave pública y Debian no tiene nuestra llave pública.

→ ssh-copy-id -p 22022 ignaciordt@192.168.56.105

Copio mi llave pública SSH al servidor utilizando el puerto SSH 22022 indicando el usuario.

```
ignaciordt@debian-ise:~$ ssh-copy-id -p 22022 ignaciordt@192.168.56.105
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ignaciordt
/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
 out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are promp
ted now it is to install the new keys
ignaciordt@192.168.56.105's password:

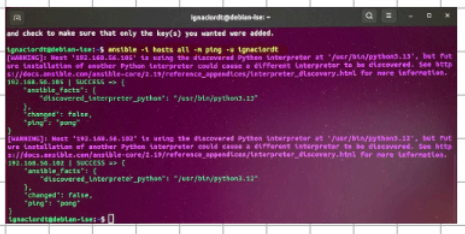
Number of key(s) added: 1

Now try logging into the machine, with: "ssh -p 22022 'ignaciordt@192.168.56.10
5'"
and check to make sure that only the key(s) you wanted were added.

ignaciordt@debian-ise:~$
```

→ ansible -i hosts all -m ping -u ignaciordt

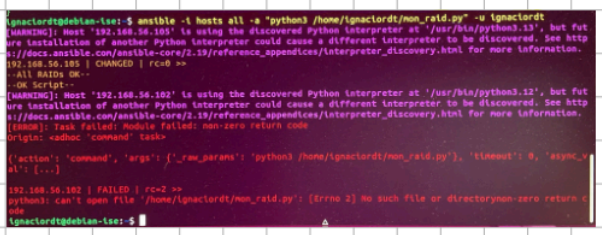
Ahora si funciona en ambos correctamente.




→ ansible -i hosts all -a "pytho..." -u ignaciordt

Ejecuta el script en cada máquina.

Ahora en Alma da error porque todavía no hemos copiado allí el script.



→ scp -P 22022 /home/ignaciordt/mon_raid.py ignaciordt@192.168.56.102:/home/ignaciordt/

Copiamos el script de Debian a Alma por el puerto 22022 para que ansible pueda ejecutarlo en ambas máquinas.

→ ansible -i hosts all -a "pytho..." -u ignaciordt

Ejecuta el script en cada máquina.

```
ignaciordt@debian-ise:~$ scp -P 22022 /home/ignaciordt/mon_raid.py ignaciordt@192.168.56.102:/home/ignaciordt/
mon_raid.py                                   100%  364   811.5KB/s   00:00

ignaciordt@debian-ise:~$ ansible -i hosts all -a "python3 /home/ignaciordt/mon_raid.py" -u ignaciordt
[WARNING]: Host '192.168.56.105' is using the discovered Python interpreter at '/usr/bin/python3.13', but
future installation of another Python interpreter could cause a different interpreter to be discovered. Se
e https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more info
rmation.
192.168.56.105 | CHANGED | rc=0 >>
--All RAIDs OK--
--OK Script--
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/python3.12', but
future installation of another Python interpreter could cause a different interpreter to be discovered. Se
e https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more info
rmation.
192.168.56.102 | CHANGED | rc=0 >>
--All RAIDs OK--
--OK Script--
ignaciordt@debian-ise:~$
```

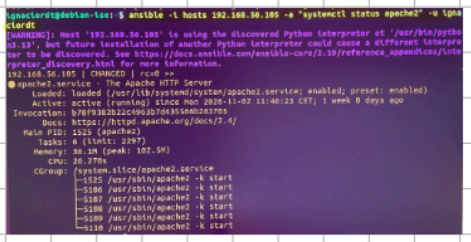
→ ¡Captura Github!

Ejercicio para punto EXTRA

**Ejercicio 2:** Usted deberá saber cómo instalar y configurar Ansible para poder hacer un ping a las máquinas virtuales de los servidores y ejecutar un comando básico (p.ej. el script de monitorización del RAID1). También debe ser consciente de la posibilidad de escribir acciones más complejas mediante playbooks escritos con YAML como, por ejemplo, asegurarse de que tenemos la última verisón instalada de httpd y que está en ejecución.

→ ansible -i hosts 192.168.56.105 -a "systemctl status apache2" -u ignaciordt

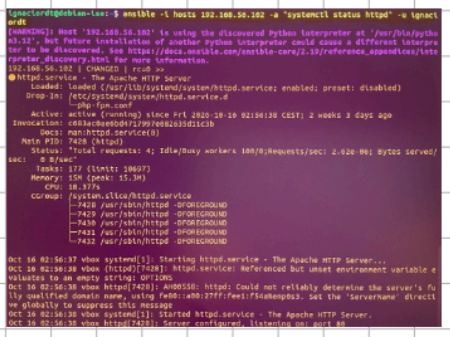
Comprobamos el estado de Apache en Debian.




→ ansible -i hosts 192.168.56.102 -a "systemctl status httpd" -u ignaciordt

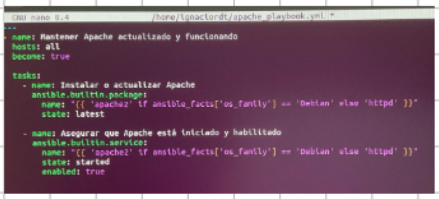
Comprobamos el estado de httpd en Alma.

En Alma el servicio Apache se llama httpd.




En Debian:

→ nano ~/apache_playbook.yml




→ Es un archivo YAML que permite automatizar la instalación, actualización y gestión de Apache en varios servidores mediante Ansible.

→ ansible -i hosts 192.168.56.105 -m command -a "whoami" -u ignaciordt --become --become-method su --ask-become-pass

En Debian, comprueba con que usuario se ejecuta el comando, solicita permisos de admin, utilizamos su - para admin.

↳ Comprueba si Ansible puede ejecutar órdenes como admin (root) en Debian, utilizando su y la contraseña de root.

```
ignaciordt@debian-ise:~$ ansible -i hosts 192.168.56.105 -m command -a "whoami" -u ignaciordt
--become --become-method su --ask-become-pass
BECOME password:
[WARNING]: Host '192.168.56.105' is using the discovered Python interpreter at '/usr/bin/pytho
n3.13', but future installation of another Python interpreter could cause a different interpre
ter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/inte
rpreter_discovery.html for more information.
192.168.56.105 | CHANGED | rc=0 >>
root
ignaciordt@debian-ise:~$
```

→ ansible -i hosts 192.168.56.102 -m command -a "whoami" -u ignaciordt --become --ask-become-pass → En Alma no ponemos su - (usa sudo)

Comprueba si Ansible puede ejecutar órdenes como admin (root) en Debian, utilizando su y la contraseña de root.

```
ignaciordt@debian-ise:~$ ansible -i hosts 192.168.56.102 -m command -a "whoami" -u ignaciordt
--become --ask-become-pass
BECOME password:
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/pytho
n3.12', but future installation of another Python interpreter could cause a different interpre
ter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/inte
rpreter_discovery.html for more information.
192.168.56.102 | CHANGED | rc=0 >>
root
ignaciordt@debian-ise:~$
```

Como hemos comprobado que sí se puede, ahora modificamos el archivo hosts.

→ nano ~/hosts

```
 GNU nano 8.4                    /home/ignaciordt/hosts *
192.168.56.105 ansible_port=22022 ansible_become_method=su
192.168.56.102 ansible_port=22022 ansible_become_method=sudo
```

— Conseguimos que Ansible sepa automáticamente cómo obtener permisos de admin en cada servidor.

→ ansible-playbook -i hosts ~/apache_playbook.yml --check --ask-become-pass

↳ Simulamos la ejecución de playbook en Debian y Alma sin realizar cambios reales.

```
ignaciordt@debian-ise:~$ ansible-playbook -i hosts ~/apache_playbook.yml --check --ask-become-pass
BECOME password:

PLAY [Mantener Apache actualizado y funcionando] ***************************************

TASK [Gathering Facts] *****************************************************************
[WARNING]: Host '192.168.56.105' is using the discovered Python interpreter at '/usr/bin/python3.13'
, but future installation of another Python interpreter could cause a different interpreter to be di
scovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.
html for more information.
ok: [192.168.56.105]
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/python3.12'
, but future installation of another Python interpreter could cause a different interpreter to be di
scovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.
html for more information.
ok: [192.168.56.102]

TASK [Instalar o actualizar Apache] ****************************************************
ok: [192.168.56.105]
changed: [192.168.56.102]

TASK [Asegurar que Apache está iniciado y habilitado] **********************************
ok: [192.168.56.105]
ok: [192.168.56.102]

PLAY RECAP *****************************************************************************
192.168.56.102             : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=
0    ignored=0
192.168.56.105             : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=
0    ignored=0

ignaciordt@debian-ise:~$
```

→ ansible-playbook -i hosts ~/apache_playbook.yml --ask-become-pass

↳ Ejecutamos playbook en Debian y Alma para mantener Apache actualizado, iniciado y habilitado en ambos servidores

```
ignaciordt@debian-ise:~$ ansible-playbook -i hosts ~/apache_playbook.yml --ask-become-pass
BECOME password:

PLAY [Mantener Apache actualizado y funcionando] ***************************************

TASK [Gathering Facts] *****************************************************************
[WARNING]: Host '192.168.56.105' is using the discovered Python interpreter at '/usr/bin/python3.13'
, but future installation of another Python interpreter could cause a different interpreter to be di
scovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.
html for more information.
ok: [192.168.56.105]
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/python3.12'
, but future installation of another Python interpreter could cause a different interpreter to be di
scovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.
html for more information.
ok: [192.168.56.102]

TASK [Instalar o actualizar Apache] ****************************************************
ok: [192.168.56.105]
changed: [192.168.56.102]

TASK [Asegurar que Apache está iniciado y habilitado] **********************************
ok: [192.168.56.105]
ok: [192.168.56.102]

PLAY RECAP *****************************************************************************
192.168.56.102             : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=
0    ignored=0
192.168.56.105             : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=
0    ignored=0

ignaciordt@debian-ise:~$
```

Ahora comprobamos si Apache sigue funcionando.

→ ansible -i hosts all -m command -a "systemctl is-active apache2" -u ignaciordt --limit 192.168.5.105 → Debian

→ ansible -i hosts all -m command -a "systemctl is-active httpd" -u ignaciordt --limit 192.168.5.102 → Alma

```
ignaciordt@debian-ise:~$ ansible -i hosts all -m command -a "systemctl is-active apache2" -u ignacio
rdt --limit 192.168.56.105
[WARNING]: Host '192.168.56.105' is using the discovered Python interpreter at '/usr/bin/python3.13'
, but future installation of another Python interpreter could cause a different interpreter to be di
scovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.
html for more information.
192.168.56.105 | CHANGED | rc=0 >>
active
ignaciordt@debian-ise:~$ ansible -i hosts all -m command -a "systemctl is-active httpd" -u ignaciord
t --limit 192.168.56.102
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/python3.12'
, but future installation of another Python interpreter could cause a different interpreter to be di
scovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.
html for more information.
192.168.56.102 | CHANGED | rc=0 >>
active
ignaciordt@debian-ise:~$
```

Vemos que sigue funcionando correctamente.

Vemos que sigue funcionando correctamente.